

1.0 Basic Concepts of Computer Networking

Introduction

When two or more computers connected together so they can exchange information, they are simply connected on network. A small network can be as simple as two computers linked together by a single cable. This course introduces you to the hardware and software needed for a network, and explains how a small network is different from larger networks and the Internet. Most networks use hubs to connect computers together. A large network may connect thousands of computers and other devices together. A wireless network connects computers without a hub or network cables but use radio communications to send data between each other. Networking allows you to share resources among a group of computer users. If you have a printer connected to your computer, you can share the printer with other computers on the network. Then instead of buying a printer for every computer, all the computers can print across the network to the printer. If you already have access to the Internet from one computer on your network, you can share that Internet connection with other computers on the network. Then all the computers on your network can browse the Web at the same time, using this single Internet connection.

1.1 Computer Network

A computer network can be defined as a collection of computers and devices interconnected by communications channels that facilitate communications among users and allow users to share resources.

1.2 Advantages and Disadvantages of computer Networks

Advantages of Computer Networks:

- i. Communication and Collaboration
- ii. Resource Sharing
- iii. Centralized Data Management
- iv. Scalability
- v. Remote Access
- vi. Automation
- vii. Enhanced Security
- viii. Reduced Costs
- ix. Improved Customer Service
- x. Real-time Monitoring and Control

Disadvantages of Computer Networks:

- i. Expensive
- ii. Virus and Malware
- iii. Management of the network
- iv. Loss of Information
- v. The system can be Hacked
- vi. Lack of Privacy
- vii. Security Difficulties
- viii. Lack of Robustness

1.3 Types of Computer Networks

Local Area Network (LAN)

A local area network is a relatively smaller network with maximum span of 10km to provide local connectivity within a building or small geographical area.

The LANs are distinguished from other kinds of networks by three characteristics:

- (i) Size
- (ii) Transmission technology, and
- (iii) Topology

Metropolitan Area Network (MAN)

Metropolitan Area Network provides regional connectivity typically within a campus or small geographical area and span in less than 50km. For example, a company can use a MAN to connect to the LANs in all of its offices throughout a city.

Wide Area Network (WAN)

Wide Area Network provides no limit of distance. A WAN provides long distance transmission of data, voice, image and video information over large geographical areas that may comprise a country, a continent or even the whole world.

1.4/1.5 Definitions of some Basic Networks Terms

Perimeter Networks: Perimeter networks are small networks that typically consist of only a few devices and are used to provide an additional layer of security between an organization's internal network and external networks like the internet.

Addressing: refers to the method used to identify devices on a network.

VLANs (Virtual Local Area Networks): VLANs are logical network segments that separate broadcast domains and improve network performance, security, and management.

Wired LANs: Wired Local Area Networks (LANs) use physical cables to connect devices within a limited geographical area, such as an office building or campus.

Wireless LANs: Wireless Local Area Networks (LANs) use radio waves to connect devices without the need for physical cables.

Leased lines: are private telecommunications circuits between two or more locations provided according to a commercial contract.

Dial-up: is a type of WAN connectivity that uses the Public Switched Telephone Network (PSTN) to establish a connection between two devices.

ISDN (Integrated Services Digital Network): is a type of digital telecommunications service that combines voice, video, and data services over a single circuit.

VPN (Virtual Private Network): is a secure network that uses encryption and other security measures to protect data transmitted over public networks.

T1 and T3: are digital transmission lines that provide high-speed data connectivity between two locations. T1 lines provide a data transfer rate of 1.544 Mbps, while T3 lines provide a data transfer rate of 44.736 Mbps

E1 and E3: are European equivalents of T1 and T3 lines, respectively. E1 lines provide a data transfer rate of 2.048 Mbps, while E3 lines provide a data transfer rate of 34.368 Mbps

DSL (Digital Subscriber Line): is a type of broadband internet connection that uses existing telephone lines to provide high-speed data connectivity.

Cable modem: is a type of broadband internet connection that uses cable television lines to provide high-speed data connectivity.

1.6 Different between client and server computers

- **Client Computer:** A client computer is a device used by end-users to access network resources and perform tasks related to their work. Clients initiate requests for services from servers and are typically smaller and less powerful than servers. **While**
- **Server Computer:** A server computer stores, processes, and distributes information for other computers on the network. Servers provide resources and services to clients, handling complex tasks like data storage, processing, and analysis. They are more powerful and have advanced hardware configurations compared to client computers.

1.7 Different between wired and wireless network

- Wired networks use physical cables such as copper wires, twisted pair, or fiber optic cables to connect devices. **while**
- Wireless networks use air as a medium to send electromagnetic or infrared waves.

Wired Networks	Wireless Networks
Uses physical cables to connect devices to the Internet or another network	Uses radio waves to connect devices to the Internet or another networks
Devices are tethered to a router	Devices can roam untethered to any wires
Higher security due to physical access control	Lower security due to radio transmissions being available to anyone within range
Higher speed and bandwidth	Lower speed and bandwidth
Dedicated connection	Shared connection
More reliable due to the absence of interference	Less reliable due to interference from other networks or wireless-enabled devices
Higher installation cost due to physical cables	Lower installation cost due to the absence of physical cables
Higher maintenance cost due to physical cables	Lower maintenance cost due to the absence of physical cables

2.0 Functions of Hardware components of computer Networks

2.1 Hardware component of Computer Networks

The hardware components of a computer network include:

- i. Network Interface Card (NIC)
- ii. Hub
- iii. Switch
- iv. Router
- v. Modem
- vi. Server
- vii. Transmission media
- viii. Repeaters
- ix. Gateway
- x. Cables

2.2 Differentiate between Hub and Switch

Hub: a hub is a device that links multiple computers and devices together. Hubs can also be referred to as repeaters or concentrators, and they serve as the center of a local area network (LAN). While

Switch: A switch is a hardware component responsible for relaying data from networks to the destination endpoint through packet switching, MAC address identification, and a multiport bridge system.

S.No.	Hub	Switch
1.	Operates at the physical layer.	Operates at the data link layer.
2.	Does not support packet switching.	Supports packet switching and has separate collision domains.
3.	Follows broadcast transmission.	Supports multicast, unicast, and broadcast transmissions.
4.	Utilizes half-duplex transmission technique.	Utilizes full-duplex transmission technique.
5.	Does not support packet filtering.	Supports packet filtering.
6.	Typically has 4 ports.	Typically contains between 24 to 28 ports.

2.3 Networks Repeaters

A repeater is a networking device that amplifies and retransmits signals to extend the reach of a transmission. It is used in both wired and wireless networks to increase the network's range, restore a damaged or weak signal, or provide access to inaccessible nodes.

Functions of Repeaters

- i. low cost
- ii. improved network performance

- iii. Enhanced signal strength
- iv. Extended network distance.

2.4 Networks Bridges

A bridge in computer networks is a device used to connect multiple Local Area Networks (LANs) together with a larger LAN.

Functions Bridges

- i. The ability to be used as a network extension.
- ii. Separate collision domains.
- iii. Increased bandwidth.
- iv. High reliability and maintainability.
- v. They can also divide the network into multiple LAN segments.
- vi. Control traffic in the network.
- vii. Interconnect two LANs with similar protocols.

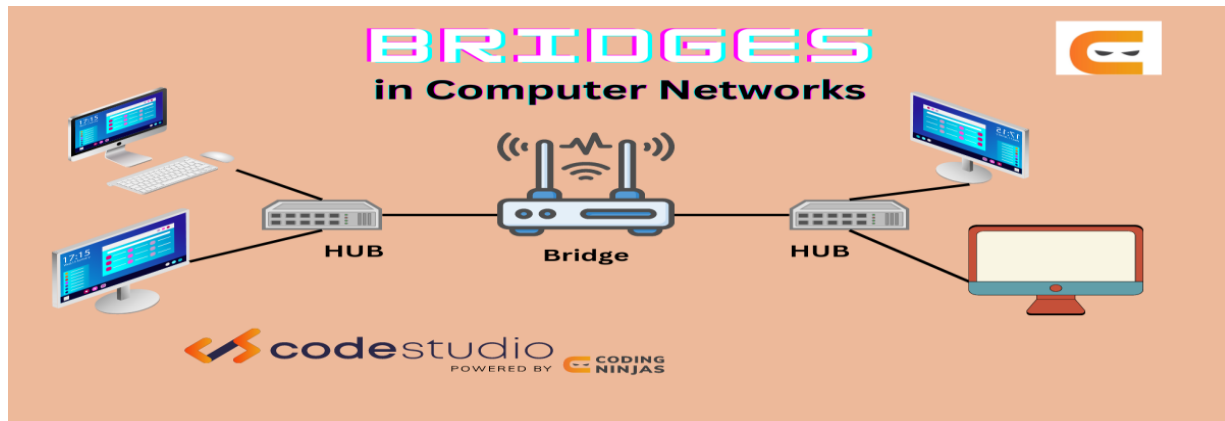


Fig 1. Bridges

Source: <http://introductiontocomputernetwork>

2.6 Networks Routers

Routers are essential devices in computer networks that receive, analyze, and forward data packets between different computer networks.

Functions of routers

- i. Traffic Management.
- ii. Connectivity.
- iii. Routing Data.
- iv. Security.
- v. Network Expansion.
- vi. Wireless Connectivity.

2.7 Network Interface Card (NIC): is a hardware component that connects a computer to a network, allowing for wired or wireless communication over the network.

Functions of NIC

- i. Transforming computer data into a network-transmittable format.

- ii. Assigning unique identifiers to data packets for source and destination identification.
- iii. Regulating data flow, ensuring correct order, and minimizing errors during transmission.
- iv. Detecting and resolving conflicts when multiple devices attempt to transmit simultaneously.
- v. Filtering unwanted traffic, identifying/responding to threats, and implementing security features like firewalls and encryption for data protection.

3.0 Network Planning and Design

3.1 Definition

Network planning and design involve the strategic process of architecting a communications network to meet the needs of subscribers and operators.

3.2 Importance of Network Planning and design

This section highlights the importance of network planning and design:

- i. Equipment and Infrastructure Identification
- ii. Ensuring that operational needs are met without sacrificing efficiency.
- iii. Improving network performance by determining optimal equipment numbers and avoiding bottlenecks.
- iv. Planning for scalability to accommodate current and future organizational needs.
- v. Minimizing network interruptions through efficient design and operation.
- vi. Ensuring network reliability.
- vii. Minimizing costs by resource optimization and reducing the need for frequent upgrades.
- viii. Ensuring network compliance with relevant regulations and standards.

3.3 Steps in designing a Network

The key steps in network planning and design include:

1. Topological design: Determining the placement of network components and how to connect them.
2. Network-synthesis: Determining the size of network components based on performance criteria like grade of service.
3. Network realization: Determining how to meet capacity requirements and ensure reliability.

3.4 Network Topology and Access point

Network topology: refers to the physical and logical arrangement of the components in a network. It describes how the various nodes (devices) in a network are connected and how data flows between them.

Types of network topology

- i. Star
- ii. Bus
- iii. Ring
- iv. Mesh
- v. Tree
- vi. Point-to-Point

Network Access methods: refer to the techniques used to control access to the shared network medium and avoid collisions.

4.0 Network connections

4.1 Types of Network connections

A peer-to-peer network: is a decentralized network architecture where participants, known as peers, interact directly with each other without the need for a central server.

Point-to-point network: is a network topology that provides a dedicated connection between two endpoints, without any intermediary devices.

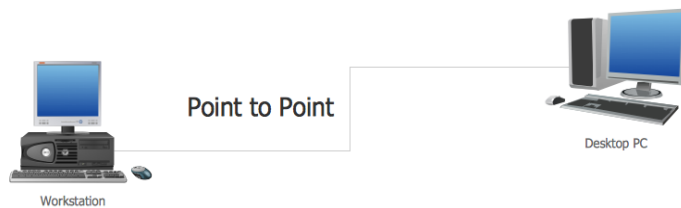


Fig. 2 Point to Point Network

Source: <http://intoructiontocomputernetwork>

A client-server network: is a distributed communications architecture where a centralized server receives and responds to requests for services and data from multiple clients.

4.2 Types of Cables termination

Termination Type	Suitable Cables
Heat Shrinkable Ends	Cables where heat can be applied to create a secure connection, providing protection against water, dust, and vibrations.
Cold Shrinkage Type	Cables where heat application is not feasible or desired.
Dry Wrap Type	Cables where a dry, non-invasive termination method is preferred.
Epoxy Resin Pouring Type	Cables that require a durable and protective seal against environmental factors.
Single Conductor Terminations	Cables with a single conductor, providing a secure connection for individual wires.
Multi-Conductor Terminations	Cables with multiple conductors, such as those used in power distribution or complex electrical systems.

Termination Type	Suitable Cables
High Voltage Terminations	High-voltage cables operating at 1kV, 10kV, 27.5kV, 35kV, 66kV, 110kV, 138kV, and 220kV, where insulation and safety are critical.

4.3 Advantages and disadvantages of each connection

(i) point-to-point connection (ii) peer-to-peer connection (iii) client server connection

Connection Type	Advantages	Disadvantages
Peer To Peer	- No central server required - Easy to set up and maintain - Highly scalable (nodes can join/leave easily)	- Less secure (each peer manages its own security) - Difficult to administer (no centralized control) - Slower performance in large networks
Point to Point	- Simple and direct communication - Low latency (direct link between devices) - Secure (only two devices involved)	- Not suitable for large networks - Limited scalability (only two devices per connection) - Requires dedicated connection
Client Server	- Centralized management (easier administration) - Scalable for large networks - Efficient resource sharing (server handles requests)	- Single point of failure (server dependency) - Expensive to set up and maintain - Less privacy (data stored centrally)

4.4 Types of Servers

- i. **Print Servers:** A print server is a specialized server that manages and controls the printing process in a network environment.
- ii. **Mail Servers:** A mail server is a computer system that processes and distributes email messages through a network.
- iii. **Web Server:** A web server is responsible for publishing websites on the internet.
- iv. **Database Server:** A database server stores and distributes databases via a network.
- v. **Web Proxy Server:** A web proxy server acts as an intermediary between clients and web servers

4.5 Server Reliability, Availability and Data Integrity

Server Reliability: Reliability refers to the probability that a server will perform its intended function without failure over a specified time period under given conditions.

Server Availability: Availability refers to the percentage of time a server is operational and accessible to users or other systems.

Data Integrity: Data integrity ensures that information stored on servers is accurate, complete, and consistent over its entire lifecycle.

5.0 Open System Interconnection (OSI) Model & TCP/IP Reference Model

5.1 OSI model

The OSI model is a conceptual framework used to describe the functions of a networking system. It is a standard model that defines a set of rules and requirements for data communication and interoperability between different devices, products, and software in a network infrastructure.

5.2 TCP/IP Reference Model

The TCP/IP (Transmission Control Protocol/Internet Protocol) Reference Model is a conceptual framework that describes the functions and protocols involved in data communication over a network.

5.3 Difference between TCP/IP and OSI Model

Parameter	TCP/IP Model	OSI Model
Number of Layers	4 layers: Application, Transport, Internet, Network Access	7 layers: Application, Presentation, Session, Transport, Network, Data Link, Physical
Approach	More horizontal, integrated approach with some overlapping between layers	Vertical, layered approach with clear boundaries between layers
Delivery Guarantee	Does not guarantee the delivery of data packets, relies on higher-level protocols for reliability	Guarantees the delivery of data packets
Replacement and Flexibility	More rigid, replacing protocols or adapting to new technologies is more challenging	Allows for easier replacement of protocols and adaptation to new technologies
Reliability	More widely adopted and proven to be more reliable in practical implementations	Considered less reliable than the TCP/IP model
History and Development	Developed by the ARPANET in the 1970s	Developed by the International Organization for Standardization (ISO)

Parameter	TCP/IP Model	OSI Model
		in the 1980s
Usage	The de facto standard for internet communication, widely used in modern networking	Primarily used as a reference model, not widely implemented in real-world networking

5.4 OSI layers and their functions

OSI Layer	Functions
Physical Layer	- Transmits message bits over a medium - Handles mechanical, electrical, procedural, and functional specifications for communication - Defines transmission mode, network topology, line configuration, data rate
Data Link Layer	- Breaks data packets into frames - Ensures error-free transmission of frames using MAC address - Provides flow control, framing, error control, physical addressing, access control
Network Layer	- Routes and forwards data packets between networks - Determines logical addressing and path selection for data transmission - Handles tasks like fragmentation and reassembly of data packets
Transport Layer	- Ensures reliable and error-free end-to-end data delivery - Provides segmentation, flow control, and error correction mechanisms - Examples include TCP and UDP
Session Layer	- Establishes, maintains, and synchronizes communication sessions between applications - Provides checkpointing, recovery, and restart capabilities for long-running sessions - Handles tasks like authentication, reconnection, session termination
Presentation Layer	- Translates, formats, and encrypts/decrypts data for the application layer - Ensures data is in a format that applications can understand - Handles data compression, encryption, character encoding
Application Layer	- Serves as the interface for network services to applications - Defines protocols and standards for specific application-level tasks - Examples include HTTP, SMTP, FTP